COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010/DATA STRUCTURES AND ALGORITHM	**SECTION:**	CPE21S1
DATE PERFORMED:	9/7/2026	**DATE SUBMITTED:**	14/7/2026
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ACTIVITY NO. _1.1__ **Basic C++ Programming**

1. PROCEDURE OUTPUT

ILO A:

Sections	Answer
Header File Declaration Section	main.cpp > main() `#include <iostream>`

Global Declaration Section	`void sumtwonum(int a, int b);`
	`bool isgreater(int a, int b);`
	`bool logicop(bool x, bool y);`

Class Declaration and Method Definition Section	NONE

Main Function

```
int main(){
int num1, num2;
bool val1, val2;

std::cout<< "number: ";
std::cin>>num1;
std::cout<< "number: ";
std::cin>>num2;

sumtwonum(num1, num2);

if (isgreater(num1,num2))
std::cout<<num1<<" is greater than "<<num2<<std::endl;
else
std::cout<<num1<< " is less than "<<num2<<std::endl;

std::cout<< "enter a value: ";
std::cin>> val1;
std::cout<< "enter a value: ";
std::cin>> val2;

if (logicop(val1,val2))
std::cout<< "Logical Operations are successful!";

return 0;
}
```

Method Definition

```
void sumtwonum(int a, int b){
    std::cout<<"sum = "<<a + b<<std::endl;

}

bool isgreater(int a, int b){
    return a>b;
}

bool logicop(bool x, bool y){
    std::cout<<"\nLOGICAL OPERATIONS"<<std::endl;
    std::cout<<"x AND y = "<<(x&&y)<<std::endl;
    std::cout<<"x OR y = "<<(x||y)<<std::endl;
    std::cout<<"NOT x = "<<(!x)<<std::endl;
    std::cout<<"NOT y = "<<(!y)<<std::endl;
    std::cout<<"x XOR y = "<<(x!=y)<<std::endl;

    return true;
}
```

ILO B:

```
PS C:\Users\TIPQC\OneDrive\Documents\CPE21S1>  & 'c:\Users\TIPQC\.vscode\extensions\ms-vscode.cpptools-1.3
2.2-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-c1v4z4t5.2sr' '--
stdout=Microsoft-MIEngine-Out-tzhziuno.zfd' '--stderr=Microsoft-MIEngine-Error-voyw3zug.gii' '--pid=Micros
oft-MIEngine-Pid-mrjhcmut.gvt' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
The shape is not a valid triangle
PS C:\Users\TIPQC\OneDrive\Documents\CPE21S1>
```

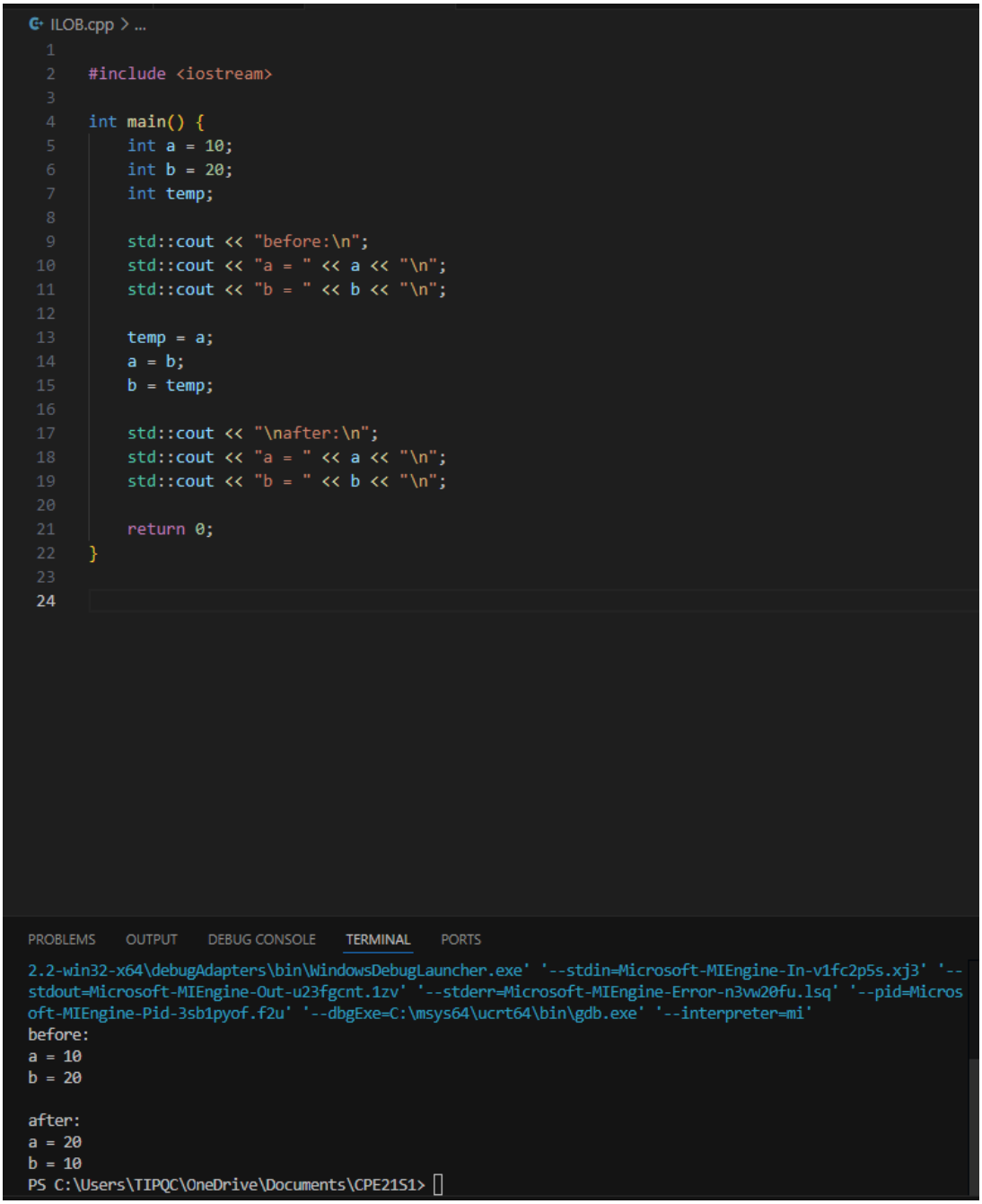
Analysis:

The code outputs “The shape is not a valid triangle” because the angles 40°, 30°, and 210° add up to 280°, not 180°. This shape does not meet the condition of a triangle.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

The supplementary activities are meant to gauge your ability in using C++. The problems below range from easy to intermediate to advanced problems. Note your difficulties after answering the problems below.

1. Create a C++ program to swap the two numbers in different variables.



```cpp
#include <iostream>

int main() {
    int a = 10;
    int b = 20;
    int temp;

    std::cout << "before:\n";
    std::cout << "a = " << a << "\n";
    std::cout << "b = " << b << "\n";

    temp = a;
    a = b;
    b = temp;

    std::cout << "\nafter:\n";
    std::cout << "a = " << a << "\n";
    std::cout << "b = " << b << "\n";

    return 0;
}
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
2.2-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-v1fc2p5s.xj3' '--stdout=Microsoft-MIEngine-Out-u23fgcnt.1zv' '--stderr=Microsoft-MIEngine-Error-n3vw20fu.lsq' '--pid=Microsoft-MIEngine-Pid-3sb1pyof.f2u' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
before:
a = 10
b = 20

after:
a = 20
b = 10
PS C:\Users\TIPQC\OneDrive\Documents\CPE21S1>
```

I found this program easy to understand because it only uses variables and simple assignments. It showed me how a temporary variable is used to swap the values of two variables without losing the original value.

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

```cpp
#include <iostream>

double kelvintofah(double kelvin) {
    return (kelvin - 273.15) * 9 / 5 + 32;
}

int main() {
    double kelvin;

    std::cout << "temperature in Kelvin: ";
    std::cin >> kelvin;

    std::cout << "temperature in Fahrenheit: "
              << kelvintofah(kelvin);

    return 0;
}
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
a = 20
b = 10
PS C:\Users\TIPQC\OneDrive\Documents\CPE21S1> ^C
PS C:\Users\TIPQC\OneDrive\Documents\CPE21S1>
PS C:\Users\TIPQC\OneDrive\Documents\CPE21S1>  & 'c:\Users\TIPQC\.vscode\extensions\ms-vscode.cpptools-1.3
2.2-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-dy1vkxs3.3uj' '--
stdout=Microsoft-MIEngine-Out-dpwgdftu.yuu' '--stderr=Microsoft-MIEngine-Error-diqhvqcn.utn' '--pid=Micros
oft-MIEngine-Pid-gnqeb0z3.po5' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
temperature in Kelvin: 100
temperature in Fahrenheit: -279.67
PS C:\Users\TIPQC\OneDrive\Documents\CPE21S1>
```

I found this program easy to understand because it only asks for one input and uses a function to convert Kelvin to Fahrenheit. It helped me learn how functions work and how formulas can be used in programming.

3. Create a C++ program that has a function that will calculate the distance between two points.

```cpp
#include <iostream>
#include <cmath>

double distance(double x1, double y1, double x2, double y2) {
    return std::sqrt(std::pow(x2 - x1, 2) + std::pow(y2 - y1, 2));
}

int main() {
    double x1, y1, x2, y2;

    std::cout << "enter the first points (x1 y1): ";
    std::cin >> x1 >> y1;

    std::cout << "enter the second points x2 y2: ";
    std::cin >> x2 >> y2;

    std::cout << "distance is= " << distance(x1, y1, x2, y2);

    return 0;
}

```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\TIPQC\OneDrive\Documents\CPE21S1>
PS C:\Users\TIPQC\OneDrive\Documents\CPE21S1>  & 'c:\Users\TIPQC\.vscode\extensions\ms-vscode.cpptools-1.3
2.2-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-cgch2yld.psz' '--
stdout=Microsoft-MIEngine-Out-tb3rroqz.jt3' '--stderr=Microsoft-MIEngine-Error-fjtpj0f0.tht' '--pid=Micros
oft-MIEngine-Pid-32zzbmy2.ber' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
enter the first points (x1 y1): 3
2
enter the second points x2 y2: 3
6
distance is= 4
PS C:\Users\TIPQC\OneDrive\Documents\CPE21S1>
```

I found this program a little more challenging than the previous ones because it uses four inputs and a mathematical formula.

4. Modify the code given in ILO B and add the following functions:
a. A function to compute for the area of a triangle
b. A function to compute for the perimeter of a triangle
c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

```cpp
#include <iostream>
#include <cmath>

class Triangle {
private:
    double totalAngle;
    double angleA, angleB, angleC;
    double sideA, sideB, sideC;

public:
    Triangle(double A, double B, double C,
             double a, double b, double c);

    void setValues(double A, double B, double C,
                   double a, double b, double c);

    bool validateTriangle();

    double computePerimeter();

    double computeArea();

    void triangleType();
};

int main() {
    Triangle set1(60, 60, 60, 5, 5, 5);

    if (set1.validateTriangle()) {
        std::cout << "The shape is a valid triangle\n";
        std::cout << "Perimeter: " << set1.computePerimeter() << std::endl;
        std::cout << "Area: " << set1.computeArea() << std::endl;
        set1.triangleType();
    }
    else {
        std::cout << "The shape is not a valid triangle\n";
    }
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\TIPQC\OneDrive\Documents\CPE21S1> ^C
PS C:\Users\TIPQC\OneDrive\Documents\CPE21S1>
PS C:\Users\TIPQC\OneDrive\Documents\CPE21S1>  & 'c:\Users\TIPQC\.vscode\extensions\ms-vscode.cpptools-1.3
2.2-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-4w4scq40.ul1' '--
stdout=Microsoft-MIEngine-Out-ndlcr4rq.qib' '--stderr=Microsoft-MIEngine-Error-meepbrcf.ypo' '--pid=Micros
oft-MIEngine-Pid-cbmev1px.53q' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
The shape is a valid triangle
Perimeter: 15
Area: 10.8253
Triangle Type: Acute-angle
```

I found this task most difficult because it uses classes, constructors, and multiple functions in one code. It took

some time to understand how the code stores the triangle's angles and side lengths before performing the calculations.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Further explanation of tasks and grammar correction**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used ChatGPT strictly to ask questions about the task (further explanation and some math formulas) and grammar correction to fully express the ideas in my head.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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